Types of Hypothesis Testing

1. What is Hypothesis Testing?

Hypothesis testing is a **statistical method** used to make decisions about a population based on sample data. It helps determine whether the observed results are **due to chance or a real effect**.

Key Concepts in Hypothesis Testing

- **Null Hypothesis (**H_0**)**: Assumes **no effect, no difference, or the status quo**.
- **Alternative Hypothesis (**H_a**)**: Represents the **research claim or change** we want to prove.
- **Significance Level (**α**)**: Common values are **0.05 or 0.01**.
- **Test Statistic**: A value (Z, t, F, χ^2) calculated from the data to determine if we reject H_0.
- **P-Value**: The probability of obtaining a test result **as extreme as or more extreme than** the observed value, assuming H_0 is true.

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2. Types of Hypothesis Tests

Hypothesis tests are categorized based on:

1. **Purpose** of the test.
2. **Number of groups being compared**.
3. **Type of data (numerical or categorical).**

A. Based on Purpose

1. One-Tailed vs. Two-Tailed Tests

Test Type	Hypothesis Structure	Example
One-Tailed Test	$H_0 : \mu \leq \mu_0$ vs. $H_a : \mu > \mu_0$ (Right-tailed) OR $H_0 : \mu \geq \mu_0$ vs. $H_a : \mu < \mu_0$ (Left-tailed)	Checking if a new drug **increases** recovery time
Two-Tailed	$H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$	Testing if a new teaching method **affects** student performance (could increase or decrease scores)

Test Type	Hypothesis Structure	Example
Test		

✅ **Example:** A company claims that **average employee salary = $50,000**.

- **One-tailed test:** Is the salary **higher** than $50,000?
- **Two-tailed test:** Is the salary **different** from $50,000?

B. Based on Number of Groups

2. Z-Test (For Large Samples, $n \geq 30$)

- Used when **population standard deviation (σ) is known**.
- **Example:** A factory claims the average weight of a product is **500g**. We sample **100 items** and use a **Z-test** to check the claim.

3. T-Test (For Small Samples, $n < 30$)

- Used when **population standard deviation (σ) is unknown**.
- **Types of t-tests**:
  1. **One-sample t-test** (comparing a sample mean to a population mean).
  2. **Two-sample t-test** (comparing means of **two independent groups**).
  3. **Paired t-test** (comparing means of **same group before and after** an event).

✅ **Example:** Testing if **10 patients** lost weight after a **new diet** using a **paired t-test**.

C. Based on Type of Data

4. Chi-Square Test (χ^2) – Categorical Data

- Used to test **relationships between categorical variables**.
- Types:
  1. **Chi-square Goodness of Fit Test** → Checks if **observed data fits an expected distribution**.
  2. **Chi-square Test of Independence** → Tests if **two categorical variables are related**.

✅ **Example:** Does **gender affect voting preference**? We survey **1000 people** and use a **Chi-square test**.

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5. ANOVA (Analysis of Variance) – Comparing More Than Two Groups

- Used when comparing **three or more group means**.
- Types:
    1. **One-way ANOVA** → Tests **one factor** (e.g., different teaching methods).
    2. **Two-way ANOVA** → Tests **two factors** (e.g., gender AND teaching methods).

✅ **Example:** Comparing **exam scores** of students using **three different teaching methods**.

D. Based on Relationship Between Variables

6. Regression Analysis – Predicting Relationships

- Used to model the relationship between **dependent and independent variables**.
- **Types:**
    - **Simple Linear Regression** (one predictor).
    - **Multiple Regression** (multiple predictors).

✅ **Example:** Predicting **house prices** based on **square footage**.

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3. Summary Table

Test Type	When to Use?	Example
Z-Test	Large samples, known σ	Checking if **average IQ** is 100
T-Test	Small samples, unknown σ	Comparing test scores of **two classes**
Chi-Square Test	Categorical variables	Do **more men than women** prefer online shopping?
ANOVA	Comparing **three or more groups**	Comparing **student scores in 3 cities**
Regression	Predicting relationships	Predicting **car price from engine size**